

SETUP & INSTALLATION GUIDE

MineOps

How to Set Up the Development Environment from Scratch

Group: Syntax Crew

Module: I3691CP

Semester 1, 2026 | Version 1.0

Prepared by: Documentation Team

1. What This Guide Is For

This guide walks any Syntax Crew member — or any new developer joining the team — through the complete process of setting up the MineOps development environment on their own computer. By the end of this guide, you will have the app running on your phone or emulator and connected to the Firebase project.

This guide assumes you are starting from a clean computer with nothing installed. Follow every step in order.

2. Prerequisites — What You Need Before Starting

2.1 Hardware Requirements

Requirement	Minimum Specification
Computer Operating System	Windows 10/11, macOS 12+, or Ubuntu 20.04+ (64-bit)
RAM	8 GB minimum — 16 GB recommended for running emulators
Storage	At least 10 GB of free disk space
Internet Connection	Required for npm installs and Firebase connection
Smartphone (for testing)	Android 8.0+ or iOS 15.0+ with the Expo Go app installed

2.2 Accounts You Need

- GitHub account — to clone the repository (ask a GitHub Manager for access to the Syntax Crew organisation).
- Expo account — free at expo.dev. Required to use EAS Build and Expo Go.
- Firebase access — ask a Firebase Engineer to add your Google account to the MineOps Firebase project.

3. Step-by-Step Installation

Step 1 — Install Node.js

Node.js is the JavaScript runtime that powers the Expo development server.

1. Go to: nodejs.org/en/download
2. Download the LTS (Long-Term Support) version for your operating system.
3. Run the installer and accept all default settings.

4. Open a terminal (Command Prompt on Windows, Terminal on Mac/Linux) and verify the install:

```
node --version  
npm --version
```

**EXPECTED
OUTPUT**

Both commands should print a version number. Node.js version 18 or higher is required. If you see an error, restart your terminal and try again.

Step 2 — Install the Expo CLI

Expo CLI is the command-line tool used to start the development server and manage the project.

Run this command in your terminal:

```
npm install -g expo-cli
```

Verify it installed correctly:

```
expo --version
```

NOTE

If you are on macOS or Linux and see a permission error, add 'sudo' before the command: `sudo npm install -g expo-cli`

Step 3 — Install Git

Git is used to download the project code and track all changes.

5. Go to: git-scm.com/downloads
6. Download and install Git for your operating system.
7. Verify the install:

```
git --version
```

Step 4 — Clone the MineOps Repository

This downloads the full project code to your computer.

8. Navigate to the folder where you want to store the project:

```
cd Documents
```

9. Clone the repository (get the correct URL from your GitHub Manager):

```
git clone https://github.com/SyntaxCrew/MineOps.git
```

10. Move into the project folder:

```
cd MineOps
```

Step 5 — Install Project Dependencies

This downloads all the JavaScript libraries the app depends on. Run this from inside the MineOps folder:

```
npm install
```

NOTE

This may take 2-5 minutes on first run. You will see a lot of text scrolling — this is normal. Wait until you see the cursor again before continuing.

Step 6 — Set Up Firebase Configuration

MineOps connects to Firebase using a configuration file that contains your project's API keys. This file is NOT stored in GitHub for security reasons — you must create it manually.

11. Ask a Firebase Engineer to send you the Firebase config values (these will include apiKey, authDomain, projectId, storageBucket, messagingSenderId, appId).
12. Inside the MineOps project folder, find the file: src/config/firebaseConfig.template.js
13. Make a copy of that file and name the copy: src/config/firebaseConfig.js
14. Open firebaseConfig.js in any text editor and replace the placeholder values with the real values from Step 1:

```
// src/config/firebaseConfig.js
import { initializeApp } from 'firebase/app';
import { getAuth } from 'firebase/auth';
import { getFirestore } from 'firebase/firestore';
import { getStorage } from 'firebase/storage';

const firebaseConfig = {
  apiKey: 'PASTE_YOUR_API_KEY_HERE',
  authDomain: 'mineops-xxxxx.firebaseio.com',
  projectId: 'mineops-xxxxx',
  storageBucket: 'mineops-xxxxx.appspot.com',
  messagingSenderId: '123456789',
```

```
  appId: '1:123456789:web:abcdef',  
};  
  
const app = initializeApp(firebaseConfig);  
export const auth = getAuth(app);  
export const db = getFirestore(app);  
export const storage = getStorage(app);
```

B7950B

Step 7 — Install Expo Go on Your Phone

15. On your Android phone: Open the Google Play Store, search for Expo Go, and install it.
16. On your iPhone: Open the App Store, search for Expo Go, and install it.
17. Open Expo Go and sign in with your Expo account (or create one at expo.dev).

Step 8 — Run the App

From inside the MineOps project folder in your terminal:

```
npx expo start
```

A QR code will appear in your terminal. Open Expo Go on your phone and scan the QR code. The MineOps app will load on your device. Any change you save in the code will automatically reload on your phone.

B7950B

4. Repository Folder Structure

Once you have cloned the repository, here is what you will find inside the MineOps folder:

Folder / File	What It Contains
src/	All application source code — this is where developers spend most of their time.
src/screens/	One file per app screen (e.g. LoginScreen.js, DashboardScreen.js).
src/components/	Reusable UI components shared across multiple screens.
src/config/	Firebase configuration file. Only firebaseConfig.template.js is in GitHub.
src/services/	Firebase service functions — all Firestore reads and writes happen here.
src/navigation/	The app navigation structure (tab navigator, stack navigator).
docs/	All documentation files — SRS, design rationale, this guide, progress reports.
designs/	Exported PNG images of all Figma prototype screens.
assets/	App icons, splash screen image, and other static image assets.
app.json	Expo configuration — app name, version, icon, splash screen settings.
package.json	Project dependencies and npm scripts.
.gitignore	List of files Git should NOT track (includes firebaseConfig.js).
README.md	Quick-start guide shown on the GitHub repository homepage.

5. Common Setup Problems and Fixes

Problem	Fix
QR code does not connect in Expo Go	Make sure your phone and computer are on the same Wi-Fi network. Or press 't' in the terminal to switch to tunnel mode.
'command not found: expo'	The Expo CLI did not install to your PATH. Try: npx expo start instead of expo start.

'npm install' gives permission errors (Mac/Linux)	Run: <code>sudo npm install -g expo-cli</code> — then enter your computer password.
Firebase connection errors on first load	Check that <code>firebaseConfig.js</code> has the correct values. Make sure your Google account has been added to the Firebase project by a Firebase Engineer.
App runs but shows a blank white screen	Check the terminal for a red error message. The most common cause is an import error in a screen file.
'Metro bundler' crashes on start	Delete the <code>node_modules</code> folder and run <code>npm install</code> again to reinstall all dependencies.

6. Making Your First Commit

Once the app is running, here is how to save your first code change to GitHub:

```
git add .  
git commit -m 'feat: Brief description of what you changed'  
git push origin main
```

Commit message rules: Always start with a prefix. Use `feat:` for new features, `fix:` for bug fixes, `docs:` for documentation changes. Never use single-word messages like `'update'` or `'fix'` — these do not count as meaningful commits and will not be assessed.

HELP

If you are stuck at any point in this guide, contact the GitHub Managers (Feliciano Nakandjibi or Frans Andreas) or the Firebase Engineers (Saara Ndweda, Ndapandula Glukua, or Gehas Imene) on the group chat.